

Introduction:

In digital electronics, combinational circuits generate outputs solely based on present inputs, whereas sequential circuits incorporate memory, allowing outputs to depend on both present inputs and previous

states. In our previous project, we implemented a combinational circuit using NAND gates to display specific alphanumeric combinations ("4J, 7H, 9P, FU, 2h, CE, 3d") on two seven segment displays. However, that system required manual input selection.

To overcome this limitation, the current project focuses on both present and previous states to design an autonomous sequential system. This is achieved using JK Flip Flops as memory elements and a 555 timer IC as a clock generator.

The system operates by transitioning through a sequence of states, each corresponding to a specific output pattern. These states are driven by clock pulses, making the display change automatically at a fixed time intervals.

The primary objectives of this project are:

① To design a state based sequential system

② To implement state transitions using JK Flip Flops

③ To integrate sequential logic with a combinational display circuit.

④ To generate a stable clock signal using a timer circuit.

⑤ To ensure smooth and continuous output cycling.

State Table

Present state			Next State			Flip Flop input Functions											
A(t)	B(t)	C(t)	A(t+1)	B(t+1)	C(t+1)	J _A	K _A	J _B	K _B	J _C	K _C	D _A	D _B	D _C	T _A	T _B	T _C
0	0	0	0	0	1	0	X	0	X	1	X	0	0	1	0	0	1
0	0	1	0	1	0	0	X	1	X	X	1	0	1	0	0	1	1
0	1	0	0	1	1	0	X	X	0	1	X	0	1	0	0	0	1
0	1	1	1	0	0	1	X	X	1	X	1	1	0	0	1	1	1
1	0	0	1	0	0	X	0	0	X	1	X	1	0	1	0	0	1
1	0	1	1	1	0	X	0	0	X	1	X	1	0	0	1	1	1
1	1	0	0	0	0	X	1	X	1	0	X	0	0	0	1	1	0
1	1	1	0	0	0	X	1	X	1	X	1	0	0	0	1	1	1

K-Map and simplified equations

$$J_A = \sum m(3) + \sum d(4, 5, 6, 7)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	0 ₀	0 ₁	1 ₃	0 ₂
$A(t)$	X ₄	X ₅	X ₇	X ₆

$$J_A = B(t)C(t)$$

$$J_B = \sum m(1, 5) + \sum d(2, 3, 6, 7)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	0 ₀	1 ₁	X ₃	X ₂
$A(t)$	0 ₄	1 ₅	X ₇	X ₆

$$J_B = C(t)$$

$$J_C = \sum m(0, 2, 4) + \sum d(1, 3, 5, 7)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	1 ₀	X ₁	X ₃	1 ₂
$A(t)$	1 ₄	X ₅	X ₇	0 ₆

$$J_C = A'(t) + B'(t)$$

$$K_A = \sum m(6, 7) + \sum d(0, 1, 2, 3)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	X ₀	X ₁	X ₃	X ₂
$A(t)$	0 ₄	0 ₅	1 ₇	1 ₆

$$\therefore K_A = B(t)$$

$$K_B = \sum m(3, 6, 7) + \sum d(0, 1, 4, 5)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	X ₀	X ₁	1 ₃	0 ₂
$A(t)$	X ₄	X ₅	1 ₇	1 ₆

$$\therefore K_B = A(t) + C(t)$$

$$K_C = \sum m(1, 3, 5, 7) + \sum d(0, 2, 4, 6)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	X ₀	1 ₁	1 ₃	X ₂
$A(t)$	X ₄	1 ₅	1 ₇	X ₆

$$\therefore K_C = 1$$

$$(1) \cdot (1) \cdot (1) \cdot (1) = 1$$

$$D_A = \sum m(3, 4, 5) \quad b \sum + (A, \omega) m \sum = \Delta$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	0 0	0 1	1 3	0 2
$A(t)$	1 4	1 5	0 7	0 6

$$\therefore D_A = A'(t)B(t)C(t) + A(t)B'(t)$$

$$D_B = \sum m(1, 2, 5)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	0 0	1 3	0 2	1 6
$A(t)$	0 4	1 5	0 7	0 6

$$\therefore D_B = A'(t)B(t)C'(t) + B'(t)C(t)$$

$$D_C = \sum m(0, 2, 4)$$

	$B'(t)C'(t)$	$B'(t)C(t)$	$B(t)C'(t)$	$B(t)C(t)$
$A'(t)$	1 0	0 1	0 3	1 2
$A(t)$	1 4	0 5	0 7	0 6

$$\therefore D_C = B'(t)C'(t) + A'(t)C'(t)$$

$$T_A = \sum m(3, 6, 7)$$

	$B'(t)C'(t)$		$B'(t)C(t)$		$B(t)C(t)$		$B(t)C'(t)$	
$A'(t)$	0	0	1	3	0	2		
$A(t)$	0	0	1	1	1	6		
	4	5	7					

$$\therefore T_A = A(t)B(t) + B(t)C(t)$$

$$T_B = \sum m(1, 3, 5, 6, 7)$$

	$B'(t)C'(t)$		$B'(t)C(t)$		$B(t)C(t)$		$B(t)C'(t)$	
$A'(t)$	0	1	3	2				
$A(t)$	4	5	7	6				

$$\therefore T_B = A(t)B(t) + C(t)$$

$$T_C = \sum m(0, 1, 2, 3, 4, 5, 7)$$

	$B'(t)C'(t)$		$B'(t)C(t)$		$B(t)C(t)$		$B(t)C'(t)$	
$A'(t)$	1	0	1	1	3	1	2	
$A(t)$	1	4	1	5	1	7	0	

$$T_C = A'(t) + B'(t) + C(t)$$

$$(A + B)'$$

Simplified equations with NAND

$$J_A = \overline{\overline{B(t)C(t)}}$$

$$J_B = \overline{\overline{C(t)}}$$

$$J_C = A'(t) + B'(t)$$

$$= \overline{\overline{A(t) + B(t)}}$$

$$= \overline{A(t) \cdot B(t)} \quad (\text{De Morgan's Law})$$

$$K_A = \overline{\overline{B(t)}}$$

$$K_B = \overline{\overline{A(t) + C(t)}} = \overline{\overline{A(t)} \cdot \overline{\overline{C(t)}}} \quad (\text{De Morgan's Law})$$

$$K_C = 1 \quad (\text{Direct VCC connection})$$

$$D_A = \overline{\overline{A(t)B(t)C(t)} + \overline{\overline{A(t)B(t)}}$$

$$= \overline{\overline{A(t)B(t)C(t)}} \cdot \overline{\overline{A(t)B(t)}} \quad (\text{De Morgan's Law})$$

$$D_B = \overline{\overline{A(t)B(t)C(t)} + \overline{\overline{B(t)C(t)}}$$

$$= \overline{\overline{A(t)B(t)C(t)}} \cdot \overline{\overline{B(t)C(t)}} \quad (\text{De Morgan's Law})$$

$$D_C = \overline{\overline{B'(t)C'(t)} + \overline{\overline{A'(t)C'(t)}}} \cdot \overline{\overline{B(t)C(t)} + \overline{\overline{A(t)C(t)}}$$

$$= C'(A' + B')$$

$$= C'(AB)' \quad (\text{De Morgan's Law})$$

$$= \overline{\overline{C} \cdot \overline{AB}}$$

$$\begin{aligned} D_B &= \overline{\overline{A(t)} B(t) \overline{C(t)} + \overline{B(t)} C} \\ &= \overline{\overline{A(t)} B(t) \overline{C(t)}} \cdot \overline{\overline{B(t)} C} \quad (\text{De Morgan's Law}) \end{aligned}$$

$$\begin{aligned} T_A &= \overline{\overline{A(t)} B(t) + B(t) C(t)} \\ &= \overline{\overline{A(t)} B(t)} \cdot \overline{B(t) C(t)} \quad (\text{De Morgan's Law}) \end{aligned}$$

$$\begin{aligned} T_B &= \overline{\overline{A(t)} B(t) + C(t)} \\ &= \overline{\overline{A(t)} B(t)} \cdot \overline{C(t)} \quad (\text{De Morgan's Law}) \end{aligned}$$

$$\begin{aligned} T_C &= \overline{\overline{A+B} \cdot \overline{A(t)} + \overline{B(t)} + C(t)} \\ &= \overline{\overline{A(t)} B(t) + C(t)} \quad (\text{De Morgan's Law}) \end{aligned}$$

$$= \overline{\overline{A(t)} B(t) + C(t)}$$

$$= \overline{\overline{A(t)} B(t)} \cdot \overline{C(t)} \quad (\text{De Morgan's Law})$$